

Faculty of Engineering - University of Ruhuna

Software Requirement Specification

SerenGuard Maritime Surveillance System

Version 1.0

14/04/2026

Version and Approvals

Version History

Version #	Date	Revised By	Reason for Change
01	08/04/2026	Dr. Chathura Seneviratne	Initial release

This document has been approved as the official Software Requirements Document for SerenGuard Maritime Surveillance System, and accurately reflects the current understanding of business requirements. Following approval of this document, requirement changes will be governed by the project's change management process.

Document Approvals

Approver Name	Project Role	Signature	Date
Dr. Chathura Seneviratne	Client		

Project Details

Project Name	SerenGuard Maritime Surveillance System
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Project End Date	
Project Sponsor	Dr. Chathura Seneviratne, HOD, EIE, University of Ruhuna
Primary Driver	Mandatory - maritime domain awareness
Division	Department of Electrical and Information Engineering
Project Manager/Dept	Keragala K.A.D.H

Overview

This document defines the high level requirements SerenGuard Maritime Surveillance System. It will be used as the basis for:

- Creating solution designs
- Developing test plans, test scripts, and test cases
- Determining project completion
- Assessing project success

Document Resources

Name	Business Unit	Role
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Thennakoon T.M.R.G.	Development Team	ML Engineer/Backend
Madhuranga M.C	Development Team	ML Engineer/Frontend
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Keragala K.A.D.H.	Development Team	Database / API Developer
Maritime Operators	End User Group	Operational Users
System Administrators	End User Group	Admin Users

Glossary of Terms

Term/Acronym	Definition
AIS	Automatic Identification System - a transponder-based system used by vessels to broadcast position, identity, speed, and heading
EO/IR	Electro-Optical/Infrared - sensor technology for visual and thermal imaging of vessels
MMSI	Maritime Mobile Service Identity - a unique 9-digit number identifying a vessel's radio equipment
SAR	Synthetic Aperture Radar - satellite-based radar imaging system
RBAC	Role-Based Access Control - access management based on user roles
Loitering	A vessel remaining in an area for an extended period without an apparent legitimate purpose
Dark Vessel	A vessel that has turned off its AIS transponder to avoid detection
VTS	Vessel Traffic Service - a maritime traffic monitoring service
PostGIS	A spatial database extension for PostgreSQL
JWT	JSON Web Token - a standard for secure authentication tokens
API	Application Programming Interface
REST	Representational State Transfer - an architectural style for APIs
MMDA	Maritime and Marine Domain Awareness

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the complete functional and non-functional requirements for the SerenGuard Maritime Surveillance System, Version 1.0. The document covers the full scope of the system, including vessel data acquisition from AIS and sensor sources, real time anomaly detection using machine learning, a role based monitoring dashboard, alert generation, and reporting capabilities. This SRS serves as the authoritative baseline for system design, development, testing, and acceptance verification.

1.2 Document Conventions

- Requirements are stated using the keyword ”shall” to indicate mandatory compliance.
- The keyword ”should” indicates a recommended but non-mandatory behaviour.
- Each functional requirement is assigned a unique identifier in the format REQ-F-XXX.
- Each non-functional requirement uses the format REQ-NF-XXX.
- User stories use the format US-XXX.
- Use cases use the format UC-XXX.
- All requirements marked [TBD] are to be resolved prior to system design.
- Priority levels are: High, Medium, Low.

1.3 Intended Audience and Reading Suggestions

Reader Type	**Sections Most Relevant**
Development Team	All sections, especially §3, §6
Project Supervisor / Client	§1, §2, §4, §5
Testers / QA	§3, §4
Database Designer	§3.3, §6
UI/UX Designer	§3.1.7, §5.3.1
Future Maintainers	§4.4, §6

1.4 Product Scope

SerenGuard is a web based maritime surveillance software system designed to support Maritime and Marine Domain Awareness (MMDA) by collecting, fusing, analysing, and presenting vessel activity data from multiple sources including AIS, EO/IR sensors, and radar. The system shall detect suspicious or anomalous vessel behaviours in real time using a combination of rule based and machine learning approaches, and shall generate timely alerts for authorised operators and analysts. SerenGuard aims to reduce response time to maritime security incidents, improve situational awareness at the control centre, and provide a structured audit trail for all surveillance activities.

1.5 References

1. AISStream WebSocket API Documentation. AISStream.io. Current, 2024. Available at: `https://aisstream.io/documentation`
2. Anomaly Detection Using Isolation Forest. DigitalOcean. 2024. Available at: `https://www.digitalocean.com/community/tutorials/anomaly-detection-isolation-forest`
3. Speed Anomaly Detection in Maritime. Frontiers in Marine Science. 2025. DOI: `https://doi.org/10.3389/fmars.2025.1547658`
4. IEEE 830-1998 – Recommended Practice for Software Requirements Specifications. IEEE. 1998.
5. IMO Maritime Security Guidelines. International Maritime Organization (IMO). Available at: `https://www.imo.org`

2 Overall Description

2.1 Product Perspective

SerenGuard is a new, self contained maritime surveillance system developed as part of the EC5406 Software Group Project at the University of Ruhuna. It is not a replacement for any existing system in the client's environment; rather, it introduces a dedicated capability for maritime domain awareness where no such automated system currently exists. The system is designed to interface with external data sources primarily the AISStream WebSocket feed and shall be capable of future integration with coastal radar and EO/IR sensor networks. The system operates as a centralised, server hosted web application accessible to authorised users through a browser based interface.

2.1.1 System Interfaces

External System	Interface Description	Purpose
AISStream WebSocket API	WebSocket over HTTPS	Real-time AIS vessel position and identity data feed
PostgreSQL + PostGIS	Internal database interface	Persistent storage of vessel tracks, alerts, and spatial data
External Maritime DB	REST API (TBD)	Integration with satellite and regional maritime data sources

2.1.2 User Interfaces

The system shall provide a web-based graphical user interface accessible via standard modern browsers (Chrome, Firefox, Edge). The interface shall include an interactive geospatial map, an alert management panel, a vessel information panel, a reporting module, and an administrative console. The interface shall be fully responsive and operable on desktop and control-centre display configurations. No specialised client software installation shall be required.

2.1.3 Hardware Interfaces

SerenGuard does not directly control physical hardware devices in the current scope. The system shall receive digitised data streams from AIS transponder networks via the AISStream WebSocket API. Integration with physical EO/IR cameras and coastal radar equipment is planned for Phase 2 and is documented as a future requirement. The server infrastructure hosting SerenGuard shall require network interface cards capable of sustained WebSocket connectivity and sufficient processing capacity to run the anomaly detection ML pipeline.

2.1.4 Software Interfaces

Software	Purpose	Interface Method
AISStream WebSocket API	Real-time AIS data ingestion	WebSocket over WSS
PostgreSQL $\geq$ 14 with PostGIS	Spatial vessel data storage	TCP/IP via pg driver
Node.js $\geq$ 18 (Express)	Backend API server	HTTP/REST
React (Frontend)	Browser-based UI	HTTP/HTTPS

2.1.5 Communications Interfaces

The system shall communicate with the AISStream data provider using the WebSocket Secure (WSS) protocol over TCP port 443. Internal communication between the frontend React application and the backend Node.js API shall use HTTPS with REST conventions over port 443. All API responses shall use the JSON data format. Authentication tokens shall be transmitted as HTTP Authorization headers using the Bearer scheme (JWT). Email notifications for alerts shall use SMTP via a configured mail server. No custom protocols are required.

2.1.6 Memory Constraints

No specific memory constraints have been imposed by the client. The system shall be deployed on server infrastructure managed by the development team. It is assumed that the deployment environment provides a minimum of 8 GB RAM and 100 GB persistent storage for the initial operational phase. These values are provisional assumptions pending confirmation from the client.

2.1.7 Operations

The system is intended for 24/7 unattended operation with human operators monitoring the dashboard during active surveillance shifts. The following operational modes are anticipated:

- Normal Monitoring Mode: Continuous AIS data ingestion, ML anomaly scoring, and dashboard refresh.
- Alert Response Mode: Triggered when an anomaly is detected; operators review and acknowledge alerts.
- Maintenance Mode: Scheduled system updates; monitoring continues in degraded mode if possible.
- Reporting Mode: Operators or analysts generate scheduled or on demand reports.

The system shall support automated daily backups of the vessel track and alert databases. Recovery procedures following failure shall restore the system to full operation within four hours.

2.1.8 Site Adaptation Requirements

For initial deployment, the following site preparation is required:

1. A server environment capable of running Node.js, PostgreSQL with PostGIS must be provisioned prior to installation.
2. A stable internet connection with minimum 10 Mbps bandwidth must be available for continuous AISStream WebSocket data ingestion.
3. The AISStream API key must be obtained and configured in the system environment file prior to activation.
4. For Phase 2 deployment incorporating local sensor data, additional network infrastructure connecting coastal radar and EO/IR stations to the central server will be required. (TBD - dependent on physical sensor procurement.)

2.2 Product Functions

The major functions SerenGuard shall perform are:

1. **Vessel Data Acquisition** - Ingest real-time AIS data streams and store vessel identity and position records.
2. **Data Fusion and Normalisation** - Correlate multi-source inputs into unified vessel track records.
3. **Anomaly Detection** - Apply statistical thresholds and Isolation Forest ML models to detect speed, route, location, and behavioural anomalies.
4. **Real-Time Monitoring Dashboard** - Display live vessel positions on an interactive geospatial map with filtering and search.
5. **Alert Generation and Management** - Generate, display, and manage alerts for detected anomalies; support acknowledgement and escalation.
6. **Role-Based Access Control** - Enforce user authentication and authorise access based on assigned roles (Admin, Operator, Analyst).
7. **Reporting** - Generate and export structured reports on vessel activity, alert history, and traffic density.
8. **API Access** - Expose REST API endpoints for integration with external maritime systems and third-party applications.
9. **Audit Logging** - Record all user actions, system events, and alert lifecycle changes.

2.3 User Classes and Characteristics

User Class	**Description**	**Technical Level**	**Primary Functions**
System Administrator	Manages user accounts, roles, permissions, and system configuration	High	User management system configuration audit log review
Operator	Front line surveillance personnel monitoring the dashboard in real time	Medium	View vessel map receive alerts acknowledge alerts annotate vessels
Analyst	Reviews historical data, generates reports, and performs deeper investigation	Medium-High	Query historical tracks generate reports review ML model outputs
API Consumer (External)	Third party systems or applications accessing SerenGuard data via REST API	Technical	Programmatic data retrieval via authenticated API calls

2.4 Operating Environment

SerenGuard shall operate as a server-hosted web application in the following environment:

- Server OS: Linux (Ubuntu 22.04 LTS or equivalent)
- Runtime: Node.js $\geq$ 18 LTS
- Databases: PostgreSQL $\geq$ 14 with PostGIS $\geq$ 3.x (spatial data and application data including RBAC)
- Client Environment: Any modern web browser (Chrome $\geq$ 110, Firefox $\geq$ 110, Edge $\geq$ 110) on Windows 10/11 or Linux desktop
- Network: Stable internet connection required for AIS data ingestion; internal LAN for control centre deployments
- Display: Minimum 1920×1080 resolution recommended for full dashboard view

2.5 Design and Implementation Constraints

Constraint	Description	Reason
C-01	The system shall use the AISStream WebSocket API as the primary AIS data source	Mandated by client requirement for AIS integration
C-02	The backend shall be implemented in Node.js using a modular architecture	Team decision; already implemented, changing would invalidate existing work
C-03	Spatial data shall be stored in PostgreSQL with the PostGIS extension	Required for geospatial query capabilities (vessel proximity, geofencing)
C-04	The frontend shall be implemented as a React web application	Team decision; ensures browser-based accessibility
C-05	All API endpoints shall be secured using JWT-based authentication	Security policy requirement
C-06	The ML anomaly detection pipeline shall use Isolation Forest as a primary algorithm	Specified in project technical design
C-07	The system shall not make autonomous decisions to intercept or act on vessels	Legal and ethical constraint human operator approval required for all actions
C-08	The system shall comply with applicable Sri Lankan maritime law and IMO security guidelines	Regulatory compliance requirement

2.6 Stakeholders

Stakeholder	Role	Interest
Dr. Chathura Seneviratne	Project Sponsor / Client	System meets maritime surveillance requirements
Maritime Operators	Primary End Users	Usable real time monitoring interface
Analysts	Secondary End Users	Accurate historical data and reporting
System Administrators	Technical End Users	Manageable user and permission system
Development Team	Builders	Clear, implementable requirements
Sri Lanka Navy / Maritime Authority	Regulatory Stakeholder	Compliance with national maritime security requirements

2.7 User Documentation

1. **User Manual** - Covers dashboard operation, alert management, vessel search, and report generation for Operators and Analysts.
2. **System Administration Guide** - Covers user management, role assignment, system configuration, and backup procedures for Administrators.
3. **API Reference Documentation** - Auto generated and manually supplemented REST API documentation (Bruno API collection provided).
4. **In-Application Help** - Contextual tooltips and help text within the dashboard interface.

2.8 Assumptions and Dependencies

Assumptions

- Reliable internet connectivity is available at the control centre for continuous AISStream WebSocket data ingestion.
- The AISStream API will remain available and its data schema will not change significantly during the project lifecycle.
- All users of the system are authorised personnel; the system is not intended for public access.
- The ML models trained on global AIS datasets will provide acceptable baseline accuracy before local Sri Lankan data is integrated in Phase 2.
- The client will provide or facilitate access to local EO/IR and radar data for Phase 2.
- Users possess basic computer literacy and familiarity with web applications.

Constraints

- The system is constrained to the maritime domain only; it shall not monitor land or airspace activities.

- Detection accuracy of ML models is dependent on the quality and volume of training data available.
- EO/IR sensor data quality is subject to environmental conditions (weather, lighting).

2.9 Apportioning of Requirements

Phase	**Sprint(s)**	**Requirements Included**
Phase 1 : MVP	Sprint 1–3	AIS data ingestion, vessel display on map, basic RBAC, alert generation (rule-based), basic dashboard
Phase 2 : ML Integration	Sprint 4–5	Isolation Forest anomaly detection, speed/route/behavioural anomaly detection, alert scoring
Phase 3 : Full System	Sprint 6	Report generation, REST API, audit logging, geofencing configuration, EO/IR integration (TBD)
Deferred (Future)	Post-project	Satellite SAR integration, mobile application, integration with VTS systems

3 Functionality Requirements

3.1 User Authentication and Role-Based Access Control

3.1.1 Description and Priority

This feature governs how users gain access to the SerenGuard system and what actions they are permitted to perform. The system enforces authentication via username and password credentials with JSON Web Token (JWT) based session management. Access to all system functionality is authorised based on the roles assigned to the authenticated user: *System Administrator*, *Operator*, or *Analyst*. This feature is a prerequisite for all other system features.

Priority	High
Benefit	9 / 9
Penalty	9 / 9
Cost	4 / 9
Risk	3 / 9

3.1.2 Stimulus/Response Sequences

Stimulus	System Response
User submits valid login credentials	System validates credentials; issues JWT token; redirects user to their role-appropriate dashboard.
User submits incorrect credentials ($\leq$5 attempts)	System returns a generic error message without revealing which field is incorrect.
User submits incorrect credentials ($>$5 consecutive attempts)	System locks the account for 15 minutes and logs the lockout event in the audit log.
User attempts to access a restricted resource	System validates JWT and RBAC permissions; returns HTTP 403 Forbidden if unauthorised.
Administrator creates a new user account	System creates the account with the assigned role and dispatches credential information to the new user.
User session token expires	System invalidates the session and redirects the user to the login page with an appropriate message.

3.1.3 Functional Requirements

- **REQ-F-001** - The system shall require all users to authenticate using a registered username and password before permitting access to any system functionality.
- **REQ-F-002** - The system shall issue a JSON Web Token (JWT) upon successful authentication. The token shall expire after a configurable session timeout period, with a default value of 60 minutes.
- **REQ-F-003** - The system shall enforce role based access control (RBAC) such that each user can only access the functionality permitted by their assigned role.

- **REQ-F-004** - The system shall lock a user account for a minimum of 15 minutes after five consecutive failed login attempts.
- **REQ-F-005** - The system shall allow the System Administrator to create, modify, deactivate, and delete user accounts.
- **REQ-F-006** - The system shall allow the System Administrator to assign one or more roles to each user account.
- **REQ-F-007** - The system shall not permit a user to elevate their own privilege level under any circumstance.
- **REQ-F-008** - When a user's session token expires or is invalidated, the system shall redirect the user to the login page and discard any cached sensitive data from the browser session.
- **REQ-F-009** - The system shall store all user passwords using a cryptographic hashing algorithm (bcrypt or equivalent with a minimum work factor of 12). Passwords shall never be stored in plaintext.

3.1.4 User Stories

US-001 - Operator Login

User Story ID	US-001
Title	Operator Login
Priority	High
Estimate	3 story points
User Story	As an *Operator*, I want to log in to the SerenGuard system using my credentials, so that I can access the real time vessel monitoring dashboard securely.
Acceptance Criteria	*Given* I am on the SerenGuard login page, *When* I enter a valid username and password and click "Login", *Then* I am redirected to the operator dashboard and a valid JWT session token is issued. *Given* I enter an incorrect password five times, *When* I attempt the sixth submission, *Then* my account is locked for 15 minutes and a lockout notification is logged.
Dependencies	REQ-F-001, REQ-F-002, REQ-F-004

US-002 - Administrator Manages User Accounts

User Story ID	US-002
Title	Administrator Manages User Accounts
Priority	High
Estimate	5 story points
User Story	As a *System Administrator*, I want to create and assign roles to new user accounts, so that I can control who accesses the surveillance system and what actions they are permitted to perform.
Acceptance Criteria	*Given* I am logged in as System Administrator, *When* I navigate to User Management, create a new account, and assign the role "Analyst", *Then* the new account appears in the user list, and the assigned user can log in with Analyst level permissions only.
Dependencies	REQ-F-005, REQ-F-006

3.1.5 Use Case Description

Use Case ID	UC-001
Use Case Name	Authenticate and Access System
Created By	Thennakoon T.M.R.G.
Last Updated By	Thennakoon T.M.R.G.
Date Created	
Date Last Updated	
Actors	Operator, Analyst, System Administrator
Description	A registered user provides credentials to the system to establish an authenticated session and gain access to role appropriate functionality.
Preconditions	1. The user has a registered account in the system. 2. The SerenGuard login page is accessible via a supported web browser.
Postconditions	1. A valid JWT session token is issued. 2. The user is redirected to their role appropriate dashboard. 3. The login event is recorded in the audit log.
Normal Course	1. User navigates to the SerenGuard login URL. 2. User enters username and password. 3. User clicks "Login". 4. System validates credentials against the RBAC database. 5. System issues a JWT token with the user's role claim. 6. System redirects the user to the appropriate dashboard.
Alternative Courses	*UC-001.AC.1* - User selects "Forgot Password": System dispatches a password reset link to the user's registered email address.

Exceptions	*UC-001.EX.1* - Invalid credentials: System returns a generic error message “Invalid username or password” without indicating which field is incorrect. *UC-001.EX.2* - Account locked: System displays “Account locked. Please try again after 15 minutes.” *UC-001.EX.3* - Database unavailable: System displays “Service temporarily unavailable.”
Includes	UC-008 (Audit Log Entry)
Priority	High
Frequency of Use	Multiple times per operational shift per user
Business Rules	*BR-001:* Only System Administrators may create or deactivate user accounts. *BR-002:* Passwords must satisfy minimum complexity requirements (minimum 8 characters, at least one uppercase letter and one digit).
Special Requirements	The login page shall load within 2 seconds. Failed login responses shall not reveal whether the username or the password is incorrect.
Assumptions	The user has been provisioned an account by the System Administrator prior to first login.
Notes and Issues	*TBD-001:* Confirm with the client whether multi factor authentication (MFA) is required.

3.1.6 Sequence Diagram

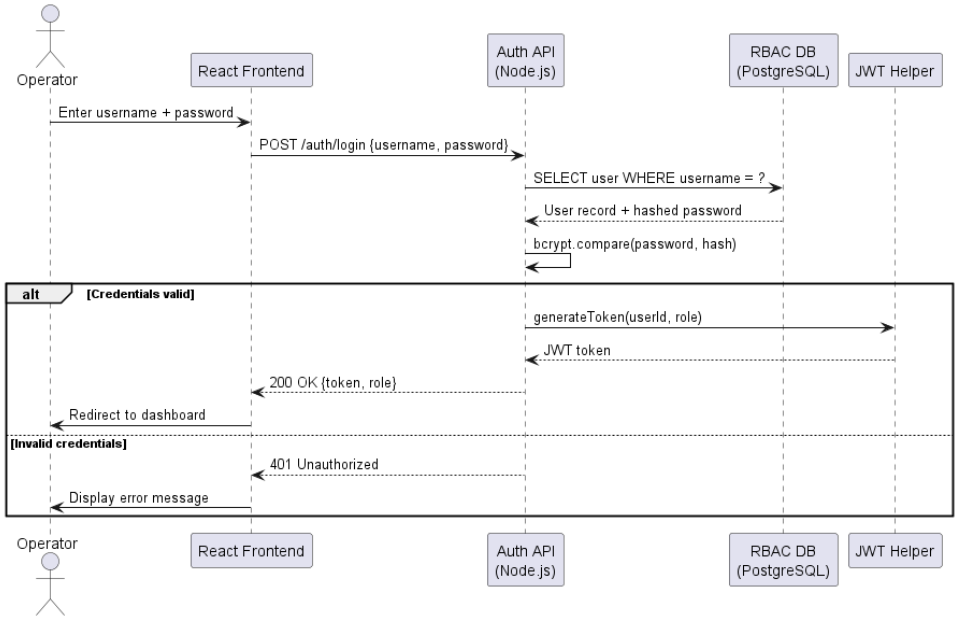


Figure 1: Sequence Diagram for User Authentication and Role Based Access Control

3.2 Vessel Data Acquisition and Ingestion

3.2.1 Description and Priority

The system shall collect real time vessel identity and position data by connecting to the AISStream WebSocket API. The Ingest Service shall parse incoming AIS messages, validate mandatory fields, enrich records with computed time range metadata, and persist them to the spatial database. Integration with radar and EO/IR sensor feeds is outside the scope of Phase 1 and is deferred to Phase 2

Priority	High
Benefit	9 / 9
Penalty	9 / 9
Cost	5 / 9
Risk	4 / 9

3.2.2 Stimulus/Response Sequences

Stimulus	System Response
System starts up	Ingest Service establishes a WebSocket connection to AISStream and begins receiving vessel messages.
Valid AIS position message received	System validates fields, computes derived metadata, and inserts a record into `vessel_positions`.
AIS message received for an MMSI not in the `vessels` table	System creates a new vessel record before inserting the position record.
AIS message received with missing mandatory fields	System discards the message and writes a validation error to the system log; no database write occurs.
WebSocket connection to AISStream is lost	System initiates reconnection attempts using exponential backoff and logs a connectivity warning.
AISStream connection cannot be restored after configured retries	System logs a critical connectivity failure and displays a visible data feed warning on the operator dashboard.

3.2.3 Functional Requirements

- **REQ-F-020** - The system shall establish and maintain a persistent WebSocket connection to the AISStream API to receive real time AIS vessel data.
- **REQ-F-021** - The system shall parse incoming AIS messages and extract the following fields at minimum: MMSI, vessel name, vessel type, latitude, longitude, speed over ground, course over ground, heading, and timestamp.
- **REQ-F-022** - The system shall validate each incoming AIS message for the presence of mandatory fields (MMSI, latitude, longitude, timestamp). Messages failing validation shall be discarded and the failure shall be written to the system error log.

- **REQ-F-023** - The system shall create a new vessel record in the `vessels` table when an AIS message is received for an MMSI not previously encountered.
- **REQ-F-024** - The system shall insert each validated position update as a new record in the `vessel_positions` table, preserving the full time series history of vessel tracks.
- **REQ-F-025** - The system shall attempt to re establish the AISStream WebSocket connection automatically if the connection is interrupted, using exponential backoff with a maximum retry interval of 60 seconds.
- **REQ-F-026** - When the AISStream connection is lost and cannot be re established, the system shall display a visible data feed status warning on the operator dashboard within 30 seconds of the interruption.
- **REQ-F-027** - The system shall not delete or overwrite historical position records. All ingested position data shall be retained for a minimum of 90 days.

3.2.4 User Stories

US-020 - Continuous AIS Data Ingestion

User Story ID	US-020
Title	Continuous AIS Data Ingestion
Priority	High
Estimate	5 story points
User Story	As a *System Administrator*, I want the system to continuously ingest AIS vessel data from the AISStream feed without manual intervention, so that the operator dashboard always reflects current vessel activity.
Acceptance Criteria	*Given* the Ingest Service is running, *When* a valid AIS message is received from AISStream, *Then* a new position record is inserted into the database within 2 seconds of receipt, and the vessel appears on the live map. *Given* the AISStream WebSocket connection drops, *When* the connection has been lost for more than 30 seconds, *Then* a data feed warning is displayed on the dashboard and automatic reconnection attempts begin.
Dependencies	REQ-F-020, REQ-F-021, REQ-F-025, REQ-F-026

3.2.5 Use Case Description

Use Case ID	UC-002
Use Case Name	Ingest Real-Time AIS Vessel Data
Created By	Madhuranga M.C.
Last Updated By	Madhuranga M.C.
Date Created	
Date Last Updated	
Actors	AISStream WebSocket API (External System), Ingest Service (System)

Description	The Ingest Service maintains a persistent WebSocket connection to AISStream and processes each received AIS message by validating, enriching, and persisting vessel identity and position data to the database.
Preconditions	1. The Ingest Service is running. 2. A valid AISStream API key is configured in the environment. 3. Network connectivity to AISStream is available.
Postconditions	1. Valid vessel position records are stored in `vessel_positions`. 2. New vessel identities are recorded in `vessels`. 3. Validation failures are written to the system error log.
Normal Course	1. Ingest Service opens WSS connection to AISStream. 2. AISStream transmits an AIS position message. 3. Ingest Service parses and validates mandatory fields. 4. If the MMSI is new, the system creates a vessel record. 5. System inserts a position record with timestamp and computed metadata. 6. Process repeats continuously for each incoming message.
Alternative Courses	*UC-002.AC.1* - Message has optional fields absent (e.g., vessel name): System inserts available fields; absent optional fields are stored as `NULL`.
Exceptions	*UC-002.EX.1* - Mandatory field missing: Message is discarded; error logged. *UC-002.EX.2* - Database write fails: System retries three times; if all retries fail, the message is written to a local fallback buffer. *UC-002.EX.3* - WebSocket disconnects: Exponential backoff reconnection is initiated; dashboard warning displayed after 30 seconds.
Includes	—
Priority	High
Frequency of Use	Continuous — every incoming AIS message, potentially thousands per minute.
Business Rules	*BR-020:* Each position record must be associated with a valid vessel MMSI. *BR-021:* No raw AIS message may be modified prior to storage.
Special Requirements	End-to-end ingestion latency (receipt to database write) shall not exceed 2 seconds.
Assumptions	*A2:* The AISStream API schema will not change without prior notice during the project period.
Notes and Issues	*TBD-007:* Data retention period is assumed at 90 days; requires client confirmation.

3.2.6 Sequence Diagram

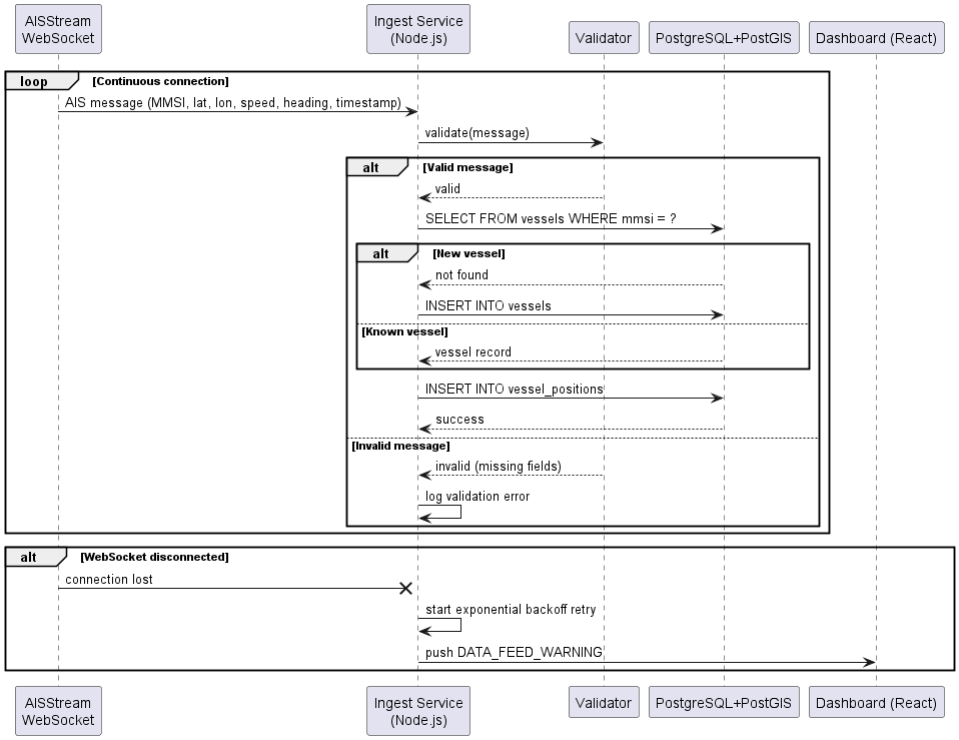


Figure 2: Sequence Diagram for Vessel Data Acquisition and Ingestion

3.3 Real-Time Vessel Monitoring Dashboard

3.3.1 Description and Priority

The system shall provide a browser-based, real time vessel monitoring dashboard that displays the current positions of all active vessels on an interactive geospatial map rendered using the Leaflet mapping library. The dashboard shall allow operators to search and filter vessels, view detailed vessel information, and monitor the AIS data feed status. In Phase 1, the dashboard retrieves updated vessel positions by polling the backend REST API at a configurable interval. Real time WebSocket push delivery to the dashboard is deferred to Phase 2

Priority	High
Benefit	9 / 9
Penalty	9 / 9
Cost	6 / 9
Risk	3 / 9

3.3.2 Stimulus/Response Sequences

Stimulus	System Response
Operator opens the dashboard in a browser	System authenticates the session, loads the Leaflet map, and renders current vessel positions retrieved via REST API.
Dashboard polling interval elapses	System issues a GET request to the vessel positions endpoint; map markers are updated with the latest positions.
Operator clicks on a vessel marker	System retrieves and displays a vessel detail panel showing MMSI, name, type, speed, heading, and last update timestamp.
Operator enters a search term (vessel name or MMSI)	System filters the displayed vessel list and highlights matching vessels on the map.
Operator applies a filter (vessel type, speed range)	System re queries the API with the filter parameters and updates the map view accordingly.
Operator zooms or pans the map	Leaflet re renders visible vessel markers in the new viewport; no additional API call is required for already loaded data.
Data feed warning is active	Dashboard displays a persistent banner indicating AIS feed interruption.

3.3.3 Functional Requirements

- **REQ-F-030** - The system shall display an interactive geospatial map using the Leaflet library, showing the current positions of all active vessels as distinct map markers.
- **REQ-F-031** - The system shall retrieve updated vessel position data by polling the backend REST API at a configurable interval, with a default value of 30 seconds.
- **REQ-F-032** - Each vessel marker on the map shall display a tooltip on hover showing the vessel's MMSI, name, and speed over ground.
- **REQ-F-033** - When an operator selects a vessel marker, the system shall display a detail panel containing: MMSI, vessel name, vessel type, flag, speed over ground, course over ground, heading, and timestamp of last position update.
- **REQ-F-034** - The system shall provide a search function allowing operators to locate vessels by MMSI or vessel name. Matching vessels shall be highlighted on the map and listed in a search results panel.
- **REQ-F-035** - The system shall allow operators to filter displayed vessels by vessel type and speed range.
- **REQ-F-036** - The dashboard shall display a visible data feed status indicator showing whether the AIS ingestion pipeline is active or interrupted.
- **REQ-F-037** - The dashboard map shall support standard Leaflet interactions: zoom, pan, and layer toggle.

- **REQ-F-038** - The dashboard shall be accessible via standard modern web browsers without requiring the installation of any additional software or browser plugins.

3.3.4 User Stories

US-030 - Operator Views Live Vessel Map

User Story ID	US-030
Title	Operator Views Live Vessel Map
Priority	High
Estimate	5 story points
User Story	As an *Operator*, I want to see all active vessels displayed on an interactive map with their current positions, so that I can maintain situational awareness of maritime activity in the monitored area.
Acceptance Criteria	*Given* I am logged in as Operator, *When* I navigate to the Dashboard, *Then* a Leaflet map loads and displays vessel markers at their current positions within 3 seconds, and the map refreshes automatically every 30 seconds.
Dependencies	REQ-F-030, REQ-F-031

US-031 - Operator Inspects Vessel Details

User Story ID	US-031
Title	Operator Inspects Vessel Details
Priority	High
Estimate	3 story points
User Story	As an *Operator*, I want to click on a vessel marker and view its full details, so that I can assess whether the vessel warrants further investigation.
Acceptance Criteria	*Given* vessel markers are displayed on the map, *When* I click on a specific marker, *Then* a detail panel appears showing MMSI, name, type, flag, speed, heading, and last update timestamp within 1 second.
Dependencies	REQ-F-033

US-032 - Operator Searches for a Specific Vessel

User Story ID	US-032
Title	Operator Searches for a Specific Vessel
Priority	Medium
Estimate	2 story points
User Story	As an *Operator*, I want to search for a vessel by its name or MMSI, so that I can quickly locate a vessel of interest on the map without manual visual scanning.
Acceptance Criteria	*Given* I am on the dashboard, *When* I type a valid MMSI or partial vessel name into the search field, *Then* matching vessels are highlighted on the map and listed in the search results panel within 1 second.
Dependencies	REQ-F-034

3.3.5 Use Case Description

Use Case ID	UC-003
Use Case Name	Monitor Vessel Activity on Real-Time Dashboard
Created By	Uduwawala W.B.W.M.R.S.H.
Last Updated By	Uduwawala W.B.W.M.R.S.H.
Date Created	
Date Last Updated	
Actors	Operator, Analyst
Description	An authenticated operator accesses the monitoring dashboard to view real time vessel positions on the Leaflet map, inspect individual vessel details, and search or filter vessels of interest.
Preconditions	1. Operator is authenticated with a valid JWT session. 2. AIS data is being ingested and vessel positions exist in the database. 3. The Leaflet map tile service is accessible.
Postconditions	1. Operator has an up to date situational picture of vessel activity. 2. No vessel data is modified by this use case.
Normal Course	1. Operator navigates to the dashboard URL. 2. System validates JWT and loads the dashboard. 3. System calls GET `/positions`. 4. Vessel markers are rendered on the Leaflet map. 5. Operator interacts with the map (zoom, pan, click). 6. Operator clicks a vessel marker; detail panel is displayed. 7. Dashboard polls the API every 30 seconds and refreshes markers.

Alternative Courses	*UC-003.AC.1* — Operator applies vessel type filter: System queries API with type parameter; only matching vessels are displayed. *UC-003.AC.2* — Operator searches by MMSI: System highlights the matching vessel and centres the map on it.
Exceptions	*UC-003.EX.1* — API returns no vessel data: Dashboard displays "No active vessels found." *UC-003.EX.2* — Map tile service unavailable: Map renders without tiles; vessel markers still displayed with coordinates. *UC-003.EX.3* — JWT expired during session: System redirects to login page.
Includes	—
Priority	High
Frequency of Use	Continuous during operational surveillance shifts
Business Rules	*BR-030:* Operators may view all vessel data but may not modify vessel records. *BR-031:* Analysts have read only access to the same dashboard view.
Special Requirements	Dashboard initial load shall complete within 3 seconds on a standard broadband connection. Map marker refresh shall not cause a full page reload.
Assumptions	*A6:* Users have access to a modern browser with JavaScript enabled.
Notes and Issues	*TBD-008:* Real time WebSocket push to the dashboard is planned for Phase 2; the current Phase 1 implementation uses API polling.

3.3.6 Sequence Diagram

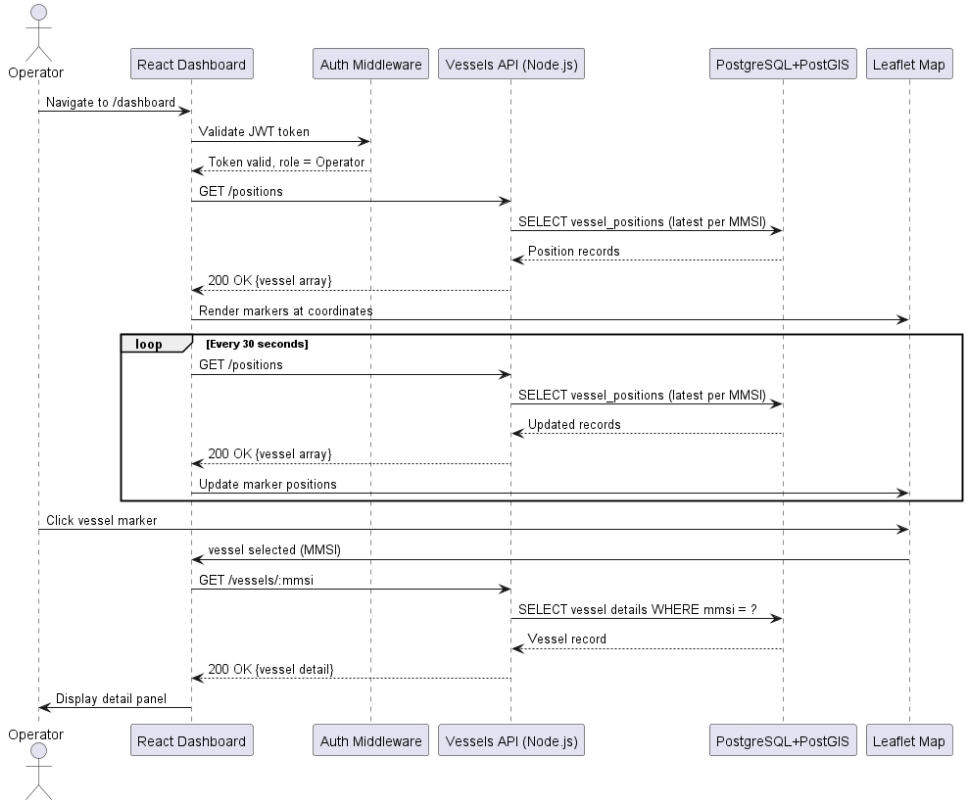


Figure 3: Sequence Diagram for Real Time Vessel Monitoring Dashboard

3.4 Anomaly Detection and Alert Generation

3.4.1 Description and Priority

The system shall automatically detect two categories of anomalous vessel behaviour in Phase 1: **speed anomalies** and **restricted area entry** (geofencing violations). Speed anomaly detection employs a two layer pipeline. The first layer applies a statistical speed threshold, defined as the 99th percentile of the speed distribution for each vessel type, computed from the training dataset and grouped by `Type` field. Records exceeding this threshold are passed to the second layer, an Isolation Forest machine learning model, which produces a continuous anomaly score. Confirmed anomalies result in alert records being persisted to the database.

Restricted area entry is detected by comparing each ingested vessel position against configured geofence polygon zones stored in the PostGIS spatial database.

The following anomaly types are deferred to Phase 2 and are not requirements of Phase 1: loitering detection, dark vessel detection, vessel type/draught mismatch detection, and unauthorised ship to ship transfer detection.

Priority	High
Benefit	9 / 9 - core intelligence capability
Penalty	9 / 9
Cost	8 / 9 - ML pipeline development
Risk	6 / 9 - model accuracy dependent on training data quality

3.4.2 Stimulus/Response Sequences

Stimulus	System Response
New AIS position record ingested	Anomaly Detection Engine retrieves the vessel type and looks up the 99th percentile speed threshold for that type.
Vessel speed exceeds the 99th percentile threshold for its type	Record is passed to the Isolation Forest model for anomaly scoring.
Isolation Forest score exceeds the configured alert threshold	System generates a speed anomaly alert with assigned severity and confidence score.
Vessel position intersects a configured geofence polygon	System generates a restricted area entry alert immediately.
Alert is generated	System writes alert record to the `alerts` table; the alert becomes visible to Operators via the alert panel at the next polling interval.

3.4.3 Functional Requirements

- **REQ-F-040** - The system shall evaluate the speed of each ingested vessel position record against the 99th percentile speed threshold for the corresponding vessel type, computed from the training dataset grouped by vessel type.
- **REQ-F-041** - When a vessel's speed exceeds the 99th percentile threshold for its type, the system shall submit the position record to the Isolation Forest anomaly detection model for scoring.
- **REQ-F-042** - When the Isolation Forest model returns an anomaly score above a configurable threshold, the system shall generate a speed anomaly alert and persist it to the `alerts` table.
- **REQ-F-043** - The system shall evaluate each ingested vessel position against all active geofence zone polygons stored in the PostGIS database.
- **REQ-F-044** - When a vessel position is detected within a restricted geofence zone, the system shall immediately generate a restricted area entry alert and persist it to the `alerts` table.
- **REQ-F-045** - Each generated alert record shall include: alert type, MMSI of the triggering vessel, detection timestamp, anomaly confidence score (where applicable), alert severity level, and initial status of "Unacknowledged".
- **REQ-F-046** - The system shall not autonomously take any operational action in response to a detected anomaly. All alert response actions shall require explicit confirmation by an authorised human operator.
- **REQ-F-047** - Speed thresholds shall be computed per vessel type as the 99th percentile of speed values from the training dataset and shall be configurable only by the System Administrator.

3.4.4 User Stories

US-040 - System Detects Speed Anomaly

User Story ID	US-040
Title	System Detects Speed Anomaly
Priority	High
Estimate	8 story points
User Story	As an *Operator*, I want the system to automatically flag vessels travelling at anomalous speeds for their vessel type, so that I can investigate potential security threats without manually reviewing every vessel track.
Acceptance Criteria	*Given* a cargo vessel is transmitting AIS data and the 99th percentile speed threshold for cargo vessels is 18 knots, *When* the vessel's recorded speed exceeds 18 knots AND the Isolation Forest model returns an anomaly score above the configured threshold, *Then* a speed anomaly alert is created in the database and is visible to the Operator on the alert panel at the next polling interval.
Dependencies	REQ-F-040, REQ-F-041, REQ-F-042

US-041 - System Detects Restricted Area Entry

User Story ID	US-041
Title	System Detects Restricted Area Entry
Priority	High
Estimate	5 story points
User Story	As an *Operator*, I want to be alerted when a vessel enters a defined restricted zone, so that I can respond to potential geofencing violations in a timely manner.
Acceptance Criteria	*Given* a restricted geofence zone is configured in the system, *When* a vessel's ingested position coordinate falls within that zone's polygon boundary, *Then* a restricted area entry alert is created in the database and appears in the Operator's alert panel at the next polling interval.
Dependencies	REQ-F-043, REQ-F-044

3.4.5 Use Case Description

Use Case ID	UC-004
Use Case Name	Detect Vessel Speed Anomaly and Generate Alert
Created By	Madhuranga M.C.
Last Updated By	Madhuranga M.C.
Date Created	
Date Last Updated	
Actors	Anomaly Detection Engine (System)

Description	Following each AIS data ingestion event, the Anomaly Detection Engine applies a two layer speed analysis pipeline. Records exceeding the 99th percentile statistical threshold are scored by the Isolation Forest model. Confirmed anomalies result in alert records being created in the database.
Preconditions	1. AIS data ingestion is active and producing position records. 2. The speed threshold lookup table (99th percentile per vessel type) is loaded. 3. The Isolation Forest model is loaded and operational.
Postconditions	1. If anomaly confirmed: alert record written to `alerts` with status "Unacknowledged." 2. If no anomaly: position record stored; no alert created. 3. All anomaly evaluation events are written to the system log.
Normal Course	1. Ingest Service delivers a validated position record to the Detection Engine. 2. Engine retrieves vessel type from `vessels`. 3. Engine looks up the 99th percentile speed threshold for the vessel type. 4. Engine compares vessel speed against the threshold. 5. Threshold exceeded: record is submitted to the Isolation Forest model. 6. Model returns an anomaly score. 7. Score exceeds the alert threshold: Engine creates alert record in `alerts`.
Alternative Courses	*UC-004.AC.1* - Vessel type not found in threshold table: Engine applies a global fallback threshold (see TBD-006). *UC-004.AC.2* - Isolation Forest score below alert threshold but above warning threshold: No alert created; a low-confidence flag is written to the system log for Analyst review.
Exceptions	*UC-004.EX.1* - ML model unavailable: System falls back to statistical threshold layer only; generates a system warning in the log; an alert is still created if the threshold is exceeded. *UC-004.EX.2* - Database write fails for alert record: System retries three times; if all retries fail, the alert is written to a local fallback file.
Includes	—
Priority	High
Frequency of Use	Executed for every ingested AIS position record; continuous operation.
Business Rules	*BR-010:* Alert thresholds are configurable only by System Administrators. *BR-011:* Alerts shall not be suppressed without explicit operator action.

Special Requirements	The full anomaly evaluation pipeline shall complete within 2 seconds of position record ingestion.
Assumptions	*A4:* The 99th percentile thresholds computed from the global training dataset provide sufficient baseline detection accuracy for Phase 1.
Notes and Issues	*TBD-005:* Finalise Isolation Forest hyperparameters (`n_estimators`, contamination rate) after training validation. *TBD-006:* Define a global fallback speed threshold for vessel types absent from the training dataset. *TBD-013:* The Isolation Forest anomaly score alert threshold value has not yet been fixed; to be determined after model training experiments.

3.4.6 Sequence Diagram

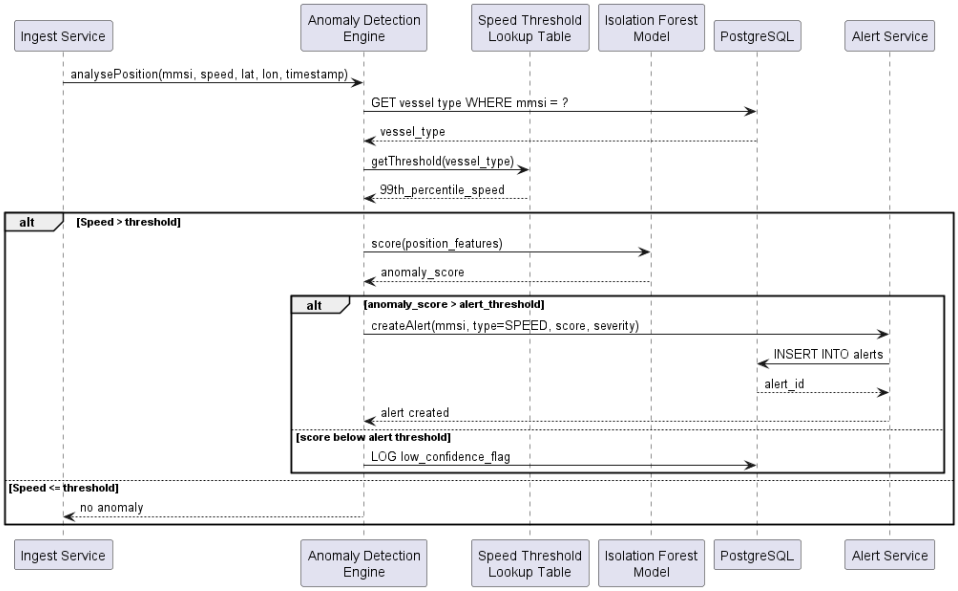


Figure 4: Anomaly Detection and Alert Generation

3.5 Alert Management

3.5.1 Description and Priority

This feature allows authorised Operators and Analysts to view, review, and acknowledge alerts generated by the anomaly detection engine. In Phase 1, alerts are retrieved via REST API polling. The system shall maintain a full alert lifecycle from creation through acknowledgement. Alert records shall be immutable and non-deletable by Operators; only System Administrators may archive alert records.

Priority	High
Benefit	9 / 9
Penalty	8 / 9
Cost	4 / 9
Risk	2 / 9

3.5.2 Stimulus/Response Sequences

Stimulus	System Response
Operator opens the Alert Panel	System retrieves all unacknowledged alerts from the alerts API endpoint and displays them sorted by severity (descending) then timestamp (descending).
Operator clicks “Acknowledge” on an alert	System updates the alert status to “Acknowledged”, records the operator’s user ID and the acknowledgement timestamp.
Operator applies a filter (type, severity, date range)	System requeries the alert API with the specified parameters and refreshes the displayed list.
Operator clicks on an alert entry	System displays the full alert detail: vessel MMSI, name, alert type, detection time, confidence score, severity, and current status.

3.5.3 Functional Requirements

- **REQ-F-050** - The system shall display all active (unacknowledged) alerts to Operators and Analysts via the Alert Panel, retrieved by polling the `GET /alerts` REST API endpoint.
- **REQ-F-051** - The system shall display each alert with the following information: alert type, vessel MMSI, vessel name, detection timestamp, severity level, confidence score (where applicable), and acknowledgement status.
- **REQ-F-052** - The system shall allow Operators to acknowledge an alert. Upon acknowledgement, the alert record shall be updated to store the acknowledging user’s ID and the acknowledgement timestamp.
- **REQ-F-053** - The system shall retain all alert records regardless of acknowledgement status. Alert records shall not be deletable by Operators or Analysts.
- **REQ-F-054** - The system shall allow filtering of the alert list by alert type, severity level, and date range.
- **REQ-F-055** - The system shall visually distinguish alert severity levels on the Alert Panel using consistent colour coding: High severity alerts shall be displayed in red, Medium in amber, and Low in blue.
- **REQ-F-056** - Only System Administrators shall have the ability to permanently delete or archive alert records.

3.5.4 User Stories

US-050 -Operator Acknowledges an Alert

User Story ID	US-050
Title	Operator Acknowledges an Alert
Priority	High
Estimate	3 story points
User Story	As an *Operator*, I want to acknowledge alerts after reviewing them, so that the alert panel reflects which events have been actioned and which still require attention.
Acceptance Criteria	*Given* an unacknowledged speed anomaly alert is displayed in the alert panel, *When* I click "Acknowledge" and confirm the action, *Then* the alert status changes to "Acknowledged", my user ID and the acknowledgement timestamp are recorded in the alert record, and the alert moves out of the active unacknowledged view.
Dependencies	REQ-F-052, REQ-F-053

3.5.5 Use Case Description

Use Case ID	UC-005
Use Case Name	Review and Acknowledge Alert
Created By	Keragala K.A.D.H.
Last Updated By	Keragala K.A.D.H.
Date Created	
Date Last Updated	
Actors	Operator; Analyst (read-only view)
Description	An authenticated Operator reviews alerts generated by the detection engine via the Alert Panel and acknowledges those that have been reviewed or actioned.
Preconditions	1. Operator is authenticated with a valid JWT session. 2. At least one alert record exists in the `alerts` table.
Postconditions	1. Alert status updated to "Acknowledged". 2. Acknowledging user ID and timestamp stored in the alert record. 3. Audit log entry created for the acknowledgement action.
Normal Course	1. Operator opens the Alert Panel. 2. System calls GET `/alerts` and retrieves unacknowledged alerts. 3. Operator reviews the alert details. 4. Operator clicks "Acknowledge." 5. System sends PATCH `/alerts/:id` with acknowledgement payload. 6. System confirms the update; alert panel refreshes to reflect the new status.

Alternative Courses	*UC-005.AC.1* — Operator filters by High severity: System displays only High severity alerts.
Exceptions	*UC-005.EX.1* — No alerts exist: Panel displays "No active alerts." *UC-005.EX.2* — API unavailable: Panel displays an error message; no data shown.
Includes	UC-008 (Audit Log Entry)
Priority	High
Frequency of Use	Multiple times per operational shift
Business Rules	*BR-004:* All alerts must be acknowledged by an Operator before being removed from the active view. *BR-032:* Analysts may view but not acknowledge alerts.
Special Requirements	The alert panel shall load within 1 second under normal load conditions.
Assumptions	—
Notes and Issues	Real time WebSocket push for new alerts is deferred to Phase 2 (TBD-008).

3.5.6 Sequence Diagram

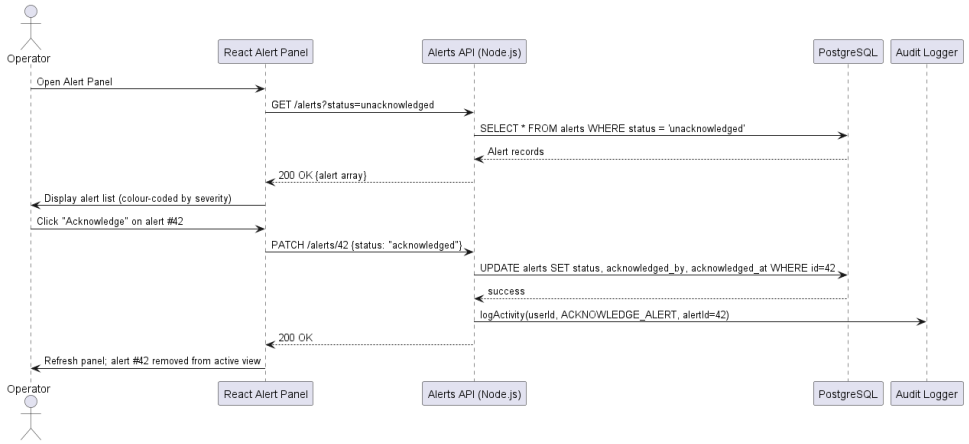


Figure 5: Sequence Diagram for Alert Management

3.6 Reporting and Data Export

3.6.1 Description and Priority

The system shall enable authorised users to generate and download structured reports covering daily vessel activity and alert history. Phase 1 supports CSV export only. PDF export and additional report types are deferred to Phase 2. Report generation is accessible to both Operators and Analysts.

Priority	Medium
Benefit	6 / 9
Penalty	4 / 9
Cost	3 / 9
Risk	2 / 9

3.6.2 Stimulus/Response Sequences

- REQ-11: The system shall export reports in standard formats

3.6.3 Functional Requirements

REQ-F-060 - The system shall provide a Daily Activity Report that summarises vessel activity for a user specified date, including: vessel MMSI, vessel name, vessel type, number of position records, minimum speed, maximum speed, first observed position, and last observed position.

REQ-F-061 - The system shall provide an Alert History Report that lists all alerts generated within a user specified date range, including: alert type, vessel MMSI, vessel name, detection timestamp, severity level, confidence score, and acknowledgement status.

REQ-F-062 - Both report types shall be exportable as CSV files downloadable directly via the browser.

REQ-F-063 - Report generation shall be accessible to Operators and Analysts. It shall not be restricted to System Administrators.

REQ-F-064 - The system shall complete CSV report generation and initiate the file download within 5 seconds for queries covering a single day of data under normal operational load.

3.6.4 User Stories

US-060 - Analyst Downloads Alert History Report

User Story ID	US-060
Title	Analyst Downloads Alert History Report
Priority	Medium
Estimate	3 story points
User Story	As an *Analyst*, I want to download a CSV report of all alerts generated over a specified date range, so that I can perform offline analysis and prepare intelligence assessments.
Acceptance Criteria	*Given* I am logged in as Analyst and navigate to the Reports module, *When* I select “Alert History”, specify a date range, and click “Download CSV”, *Then* a CSV file is downloaded containing one row per alert with all required fields, and the download begins within 5 seconds.
Dependencies	REQ-F-061, REQ-F-062, REQ-F-064

US-061 - Operator Downloads Daily Activity Report

User Story ID	US-061
Title	Operator Downloads Daily Activity Report
Priority	Medium
Estimate	3 story points
User Story	As an *Operator*, I want to download a CSV report of vessel activity for a specific date, so that I can review the day's maritime traffic and identify patterns for shift handover.
Acceptance Criteria	*Given* I am logged in as Operator and navigate to the Reports module, *When* I select "Daily Activity", choose a date, and click "Download CSV", *Then* a CSV file is downloaded containing one summary row per active vessel, with all required fields, within 5 seconds.
Dependencies	REQ-F-060, REQ-F-062, REQ-F-064

3.6.5 Use Case Description

Use Case ID	UC-006
Use Case Name	Generate and Download CSV Report
Created By	Keragala K.A.D.H.
Last Updated By	Keragala K.A.D.H.
Date Created	
Date Last Updated	
Actors	Operator, Analyst
Description	An authenticated user selects a report type and specifies query parameters (date or date range). The system executes the query, compiles the result set, and delivers a CSV file as a browser download.
Preconditions	1. User is authenticated as Operator or Analyst. 2. Relevant data exists in the database for the requested period.
Postconditions	1. A CSV file is delivered to the user's browser. 2. Report generation event is logged in the audit log.
Normal Course	1. User navigates to the Reports module. 2. User selects report type (Daily Activity or Alert History). 3. User specifies the date or date range and clicks "Download CSV". 4. System queries the database for the requested data. 5. System formats results into a CSV structure. 6. System returns the CSV file as an HTTP attachment response. 7. Browser initiates the file download.
Alternative Courses	*UC-006.AC.1* - No data exists for the selected period: System returns an empty CSV with header row only and displays an informational message.

Exceptions	*UC-006.EX.1* - Query execution exceeds 5 seconds: System displays a timeout error and advises the user to narrow the date range.
Includes	UC-008 (Audit Log Entry)
Priority	Medium
Frequency of Use	Daily or as required per operational shift
Business Rules	*BR-040:* Reports may not include data beyond the user's permitted role scope.
Special Requirements	CSV download shall begin within 5 seconds for single-day queries.
Assumptions	—
Notes and Issues	PDF export is deferred to Phase 2.

3.6.6 Sequence Diagram

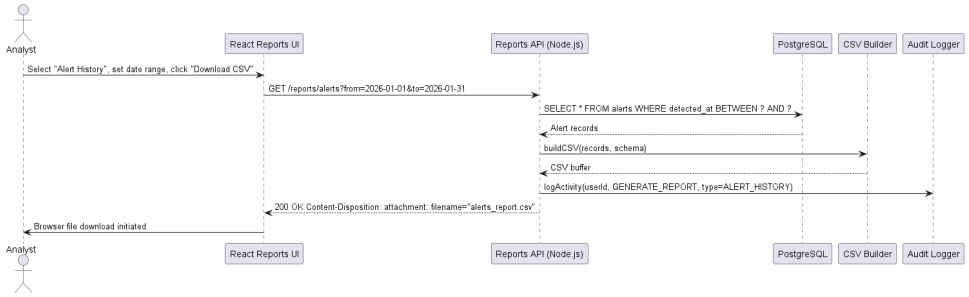


Figure 6: Sequence Diagram for Reporting and Data Export

3.7 REST API Access

3.7.1 Description and Priority

The system shall expose a documented public REST API providing authenticated access to AIS vessel position data and vessel identity data. The API is intended for consumption by external maritime systems and authorised third-party applications. All endpoints shall require JWT authentication. API documentation is provided in Bruno API collection format; an OpenAPI/Swagger specification shall be produced before system delivery.

Priority	Medium
Benefit	7 / 9
Penalty	5 / 9
Cost	4 / 9
Risk	3 / 9

3.7.2 Stimulus/Response Sequences

- REQ-12: The system shall expose REST APIs

3.7.3 Functional Requirements

REQ-F-070 - The system shall expose a documented REST API providing authenticated access to AIS vessel position data and vessel identity data.

REQ-F-071 - All API endpoints shall require a valid JWT bearer token in the HTTP Authorization header. Requests without a valid token shall receive an HTTP 401 Unauthorized response.

REQ-F-072 - The API shall return all data responses in JSON format using a consistent response envelope structure.

REQ-F-073 - The API shall enforce role based access control such that API consumers can only access endpoints permitted by their assigned role.

REQ-F-074 - The API shall provide at minimum the following endpoints: `POST /auth/login`, `GET /vessels`, `GET /vessels/:mmsi`, `GET /positions`, and `GET /alerts` (role-restricted).

REQ-F-075 - API documentation shall be maintained in Bruno API collection format. An OpenAPI/Swagger specification shall be produced and delivered with the system.

3.7.4 User Stories

US-070 — External System Retrieves Vessel Positions

User Story ID	US-070
Title	External System Retrieves Vessel Positions
Priority	Medium
Estimate	5 story points
User Story	As an *API Consumer* (external maritime authority system), I want to retrieve current vessel position data via a REST API, so that I can integrate SerenGuard data into my own command-and-control platform.
Acceptance Criteria	*Given* I have obtained a valid JWT by authenticating at `POST /auth/login`, *When* I send `GET /positions` with the Bearer token in the Authorization header, *Then* I receive a 200 OK response containing a JSON array of the latest position record for each active vessel, within 1 second.
Dependencies	REQ-F-070, REQ-F-071, REQ-F-072

3.7.5 Use Case Description

Use Case ID	UC-007
Use Case Name	Authenticate and Retrieve Data via REST API
Created By	Thennakoon T.M.R.G.
Last Updated By	Thennakoon T.M.R.G.
Date Created	
Date Last Updated	
Actors	API Consumer (External System)

Description	An authorised external system authenticates with the SerenGuard REST API and retrieves vessel position or identity data for integration into its own platform.
Preconditions	1. The external system has been issued a SerenGuard API account by the System Administrator. 2. The SerenGuard REST API server is running and accessible over HTTPS.
Postconditions	1. Requested data is returned in JSON format. 2. API access event is recorded in the audit log.
Normal Course	1. External system sends `POST /auth/login` with credentials. 2. System validates credentials and returns a JWT. 3. External system sends `GET /positions` with Bearer token. 4. System validates JWT and RBAC permissions. 5. System queries the database and returns the result in JSON.
Alternative Courses	*UC-007.AC.1* - Client requests vessel detail by MMSI: Client sends `GET /vessels/:mmsi`; system returns the matching vessel record.
Exceptions	*UC-007.EX.1* - JWT invalid or expired: System returns HTTP 401. *UC-007.EX.2* - Endpoint outside role permissions: System returns HTTP 403. *UC-007.EX.3* - Requested MMSI not found: System returns HTTP 404.
Includes	UC-008 (Audit Log Entry)
Priority	Medium
Frequency of Use	Varies by external system integration
Business Rules	*BR-007:* No unauthenticated API endpoint shall expose vessel or alert data.
Special Requirements	API responses shall complete within 1 second for standard queries under normal load.
Assumptions	The external system is responsible for securely storing its credentials and JWT tokens.
Notes and Issues	—

3.7.6 Sequence Diagram

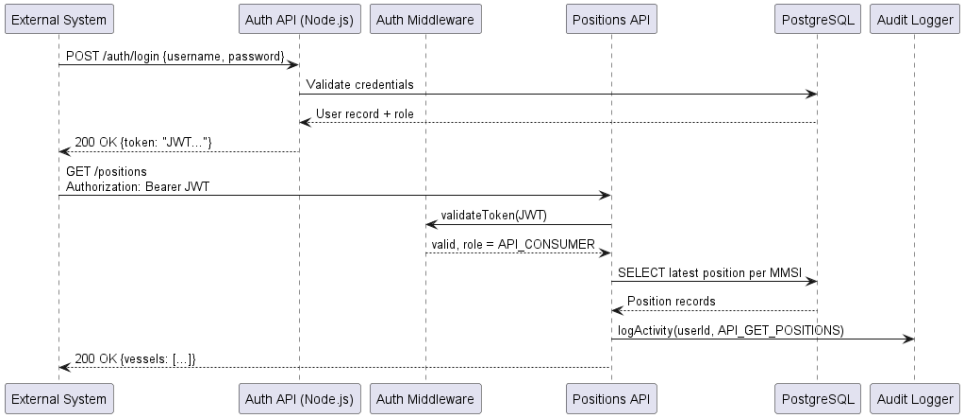


Figure 7: Sequence Diagram for REST API Access

3.8 Audit Logging and Activity Tracking

3.8.1 Description and Priority

The system shall maintain a comprehensive, immutable audit log of all user actions, authentication events, alert lifecycle changes, and system configuration modifications. Audit log records shall be written to the `activity_logs` table and shall be queryable by System Administrators. No user role, including System Administrator, shall be permitted to modify or delete audit log records. This feature is required for security compliance and operational accountability.

Priority	High
Benefit	8 / 9
Penalty	7 / 9
Cost	3 / 9
Risk	2 / 9

3.8.2 Stimulus/Response Sequences

- REQ-13: The system shall log all user actions

3.8.3 Functional Requirements

REQ-F-080 - The system shall record an audit log entry for every authentication event, including successful logins, failed login attempts, and account lockouts.

REQ-F-081 - The system shall record an audit log entry for every alert acknowledgement action, including the user ID, alert ID, action type, and timestamp.

REQ-F-082 - The system shall record an audit log entry for every user management action performed by a System Administrator, including account creation, modification, deactivation, and role assignment changes.

REQ-F-083 - The system shall record an audit log entry whenever a System Administrator modifies system configuration, including speed threshold values and geofence zone definitions.

REQ-F-084 - Audit log records shall not be modifiable or deletable by any user role, including System Administrators.

REQ-F-085 - Each audit log entry shall include at minimum: timestamp, user ID, action type, affected entity type, affected entity ID, and originating IP address.

REQ-F-086 - System Administrators shall be able to query and view audit log records filtered by user ID, action type, and date range.

3.8.4 User Stories

US-080 — Administrator Reviews Audit Log

User Story ID	US-080
Title	Administrator Reviews Audit Log
Priority	High
Estimate	3 story points
User Story	As a *System Administrator*, I want to view a filterable audit log of all user actions and system events, so that I can investigate security incidents and ensure accountability for all operations performed within the system.
Acceptance Criteria	*Given* I am logged in as System Administrator, *When* I navigate to the Audit Log section and apply a filter by user and date range, *Then* the system displays all matching audit log entries with timestamp, user, action type, and affected entity details within 2 seconds.
Dependencies	REQ-F-085, REQ-F-086

3.8.5 Use Case Description

Use Case ID	UC-008
Use Case Name	Record and Review Audit Log Entry
Created By	Madhuranga M.C.
Last Updated By	Madhuranga M.C.
Date Created	
Date Last Updated	
Actors	System (automated logging); System Administrator (review)
Description	The system automatically records an audit log entry whenever a qualifying user action or system event occurs. System Administrators may query and review the audit log using filter criteria.
Preconditions	1. A qualifying system event or user action has occurred. 2. For review: Administrator is authenticated with a valid JWT session.

Postconditions	1. An immutable audit log entry has been written to `activity_logs`. 2. For review: Matching log entries displayed to the Administrator.
Normal Course (Logging)	1. A qualifying event occurs (e.g., user login, alert acknowledgement). 2. The responsible module calls the Audit Logger service. 3. Logger writes entry to `activity_logs` with all required fields.
Normal Course (Review)	1. Administrator navigates to the Audit Log section. 2. Administrator specifies filter criteria (user, action type, date range). 3. System queries `activity_logs` with the specified filters. 4. System displays matching entries in a paginated table.
Alternative Courses	*UC-008.AC.1* - No entries match filter criteria: System displays "No log entries found for the specified criteria."
Exceptions	*UC-008.EX.1* - Log write fails (e.g., database unavailable): System writes the entry to a local fallback log file and raises a system alert. *UC-008.EX.2* - Administrator attempts to delete a log entry: System returns HTTP 403 Forbidden; operation is not permitted.
Includes	—
Priority	High
Frequency of Use	Logging: triggered by every qualifying event; continuous. Review: on demand by Administrator.
Business Rules	*BR-050:* Audit log records are immutable; no role may modify or delete them. *BR-051:* All qualifying events must produce an audit entry before returning a response to the caller.
Special Requirements	Audit log write operations shall not block or delay the primary operation being logged by more than 100 milliseconds.
Assumptions	The `activity_logs` table has sufficient storage for at least 12 months of operational logging.
Notes and Issues	—

3.8.6 Sequence Diagram

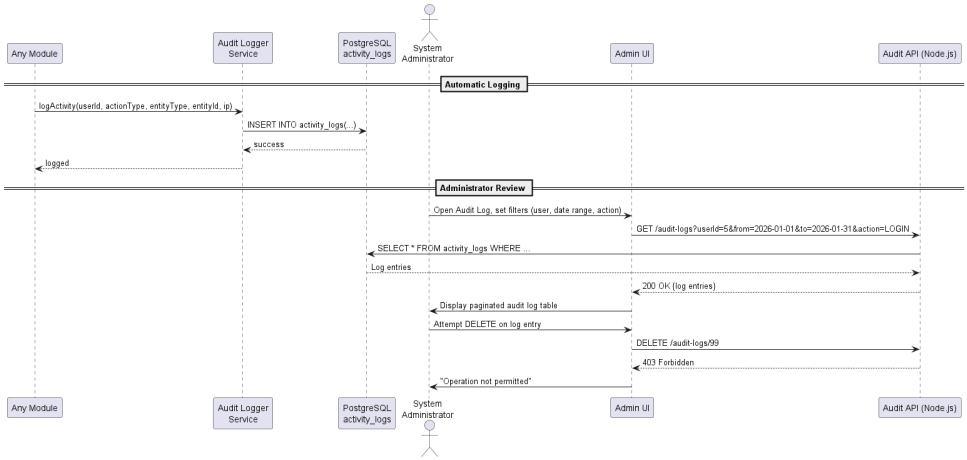


Figure 8: Sequence Diagram for Audit Logging and Activity Tracking

4 Non-functional Requirements

4.1 Performance Requirements

- **REQ-NF-001:** The system shall ingest and process AIS position updates with an end-to-end latency not exceeding 2 seconds from data receipt to dashboard display under normal operational load.
- **REQ-NF-002:** The system shall support concurrent monitoring of a minimum of 5,000 active vessel tracks without degradation in response time.
- **REQ-NF-003:** All REST API responses shall complete within 1 second for standard queries under normal load conditions.
- **REQ-NF-004:** The anomaly detection pipeline shall complete analysis of each incoming AIS record within 2 seconds of receipt.
- **REQ-NF-005:** The dashboard map shall refresh vessel position markers within 3 seconds of receiving updated position data.

4.2 Safety Requirements

- **REQ-NF-010:** The system shall not autonomously command, intercept, or take any physical action with respect to any vessel; all response actions shall require explicit authorisation by a human operator.
- **REQ-NF-011:** The system shall clearly distinguish between confirmed alerts and tentative anomaly flags on the operator dashboard to prevent inappropriate escalation.
- **REQ-NF-012:** The system shall display a prominent system health indicator on the dashboard; if the AIS data feed is interrupted, the system shall display a visible warning within 30 seconds.

4.3 Security Requirements

- **REQ-NF-020:** The system shall authenticate all users using credentials validated against the RBAC database before permitting access to any functionality.
- **REQ-NF-021:** The system shall encrypt all data in transit using TLS 1.2 or higher.
- **REQ-NF-022:** The system shall enforce role-based access control such that no user can access functionality beyond the permissions assigned to their role.
- **REQ-NF-023:** The system shall store all user passwords using bcrypt hashing with a minimum work factor of 12.
- **REQ-NF-024:** The system shall invalidate all active session tokens upon a user account being deactivated by an administrator.
- **REQ-NF-025:** The system shall log all authentication events, including successful logins, failed attempts, and account lockouts, in the audit log.
- **REQ-NF-026:** The system shall comply with applicable IMO maritime cybersecurity guidelines (MSC-FAL.1/Circ.3).

4.4 Software Quality Attributes

	Characteristic	Priority	Justification for SerenGuard
1	Correctness	High	Incorrect vessel positions or missed alerts have direct security consequences
2	Reliability	High	24/7 operation — system failure during a surveillance shift is unacceptable
3	Availability	High	Minimum 99.9% uptime required
4	Security / Integrity	High	Sensitive maritime data; access must be strictly controlled
5	Usability	High	Operators may be under stress; interface must be immediately interpretable
6	Maintainability	Medium	Modular architecture enables addition of new sensors and ML models
7	Interoperability	Medium	Must interface with AISStream; future VTS integration
8	Scalability	Medium	Must support additional coastal stations in Phase 2
9	Testability	Medium	ML pipeline and alert logic must be verifiable
10	Portability	Low	Designed for control centre deployment; not required to run on mobile OS
11	Efficiency	Medium	Processing thousands of AIS records in real time
12	Auditability	High	All user actions and system events must be logged for compliance

4.5 Business Rules

- **BR-001:** Only System Administrators may create or deactivate user accounts.
- **BR-001:** Alert thresholds (speed limits, geofence boundaries, loitering duration) may only be modified by System Administrators.
- **BR-001:** Operators may acknowledge alerts but may not delete alert records.
- **BR-001:** All generated alerts must be acknowledged by an authorised Operator before they are removed from the active alert panel.
- **BR-001:** The system shall retain vessel track data for a minimum of 90 days before archiving.
- **BR-001:** No physical or operational action against a vessel may be taken through the system; SerenGuard is a decision-support tool only.
- **BR-001:** API access requires authentication; no public unauthenticated endpoints shall expose vessel or alert data.

5 Other Requirements

5.1 Online Documentation

The system shall include contextual help tooltips on all major dashboard components. An online user manual shall be accessible from within the application via a Help menu. API documentation shall be provided in OpenAPI/Swagger format.

5.2 Purchased Components

The AISStream WebSocket API subscription constitutes an externally procured service. The development team shall ensure compliance with AISStream's terms of service. All other components (Node.js, PostgreSQL, PostGIS, React) are open-source with no licensing fees.

5.3 Interfaces

5.3.1 User Interfaces

Web-based dashboard; no native app. Resolution minimum 1920×1080. Consistent use of colour coding for alert severity (Red=High, Amber=Medium, Green=Low). All interactive elements shall be keyboard-accessible.

5.3.2 Hardware Interfaces

No direct hardware control in Phase 1. Phase 2 EO/IR interface is TBD.

5.3.3 Software Interfaces

AISStream WSS API, PostgreSQL+PostGIS, Node.js Express, React.

5.3.4 Communications Interfaces

WSS (AIS), HTTPS/REST (API), SMTP (email alerts), WebSocket (real-time dashboard push).

5.4 Licensing Requirements

All open-source libraries used in SerenGuard shall comply with their respective licences. Libraries under GPL must not be incorporated into proprietary components. The AISStream API must be used in accordance with its published terms of service.

5.5 Legal Notices

SerenGuard is developed as a university academic project under the supervision of the Faculty of Engineering, University of Ruhuna. All rights reserved by the development team and the university. The system shall not be deployed in a production maritime security environment without review and approval by competent maritime authorities.

5.6 Applicable Standards

- IEEE 830-1998 - Recommended Practice for Software Requirements Specifications
- IMO MSC-FAL.1/Circ.3 - Guidelines on Maritime Cyber Risk Management
- ITU-R M.1371 - Technical Characteristics for AIS
- OWASP Top 10 - Web Application Security Standards

6 System Architecture

6.1 Functional Architecture

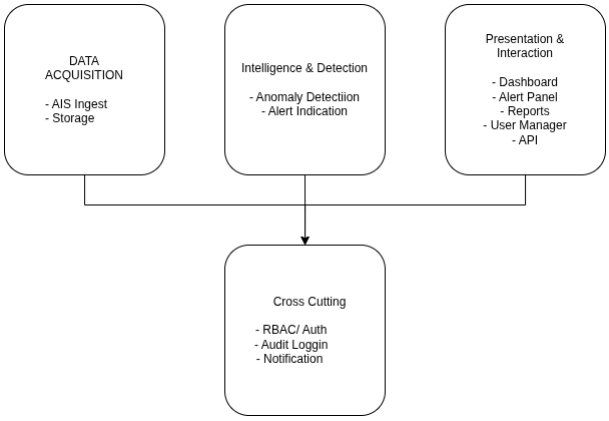


Figure 9: SerenGuard Functional Architecture

6.2 Logical Architecture

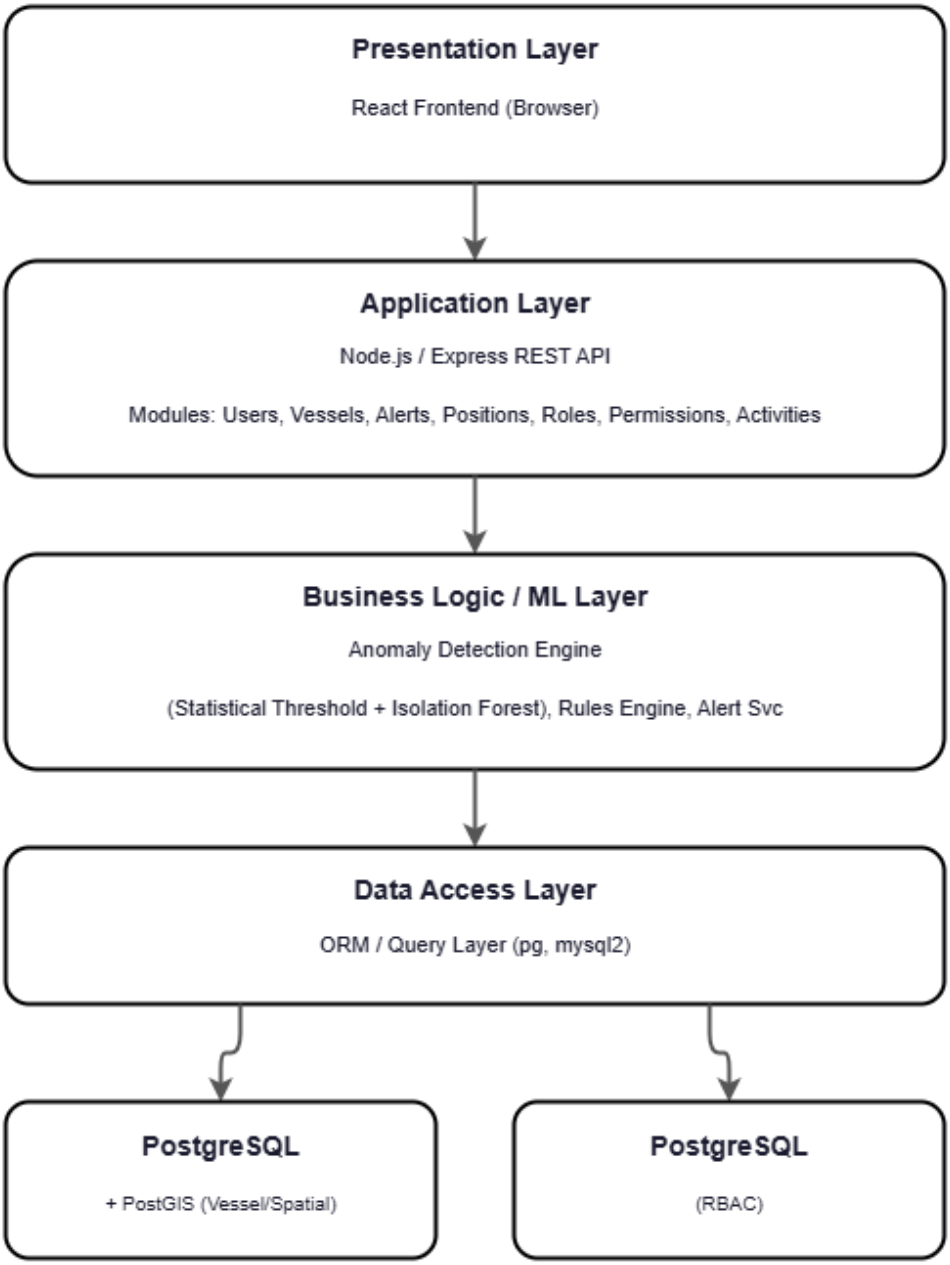


Figure 10: SerenGuard Logical Architecture

6.3 Component Architecture

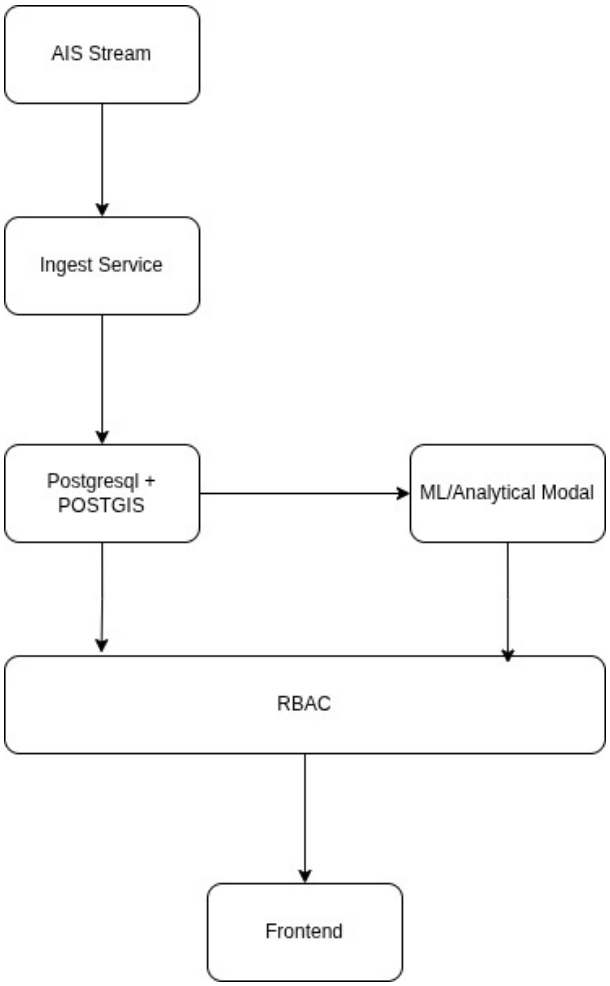


Figure 11: SerenGuard Component Architecture

6.4 Physical Architecture

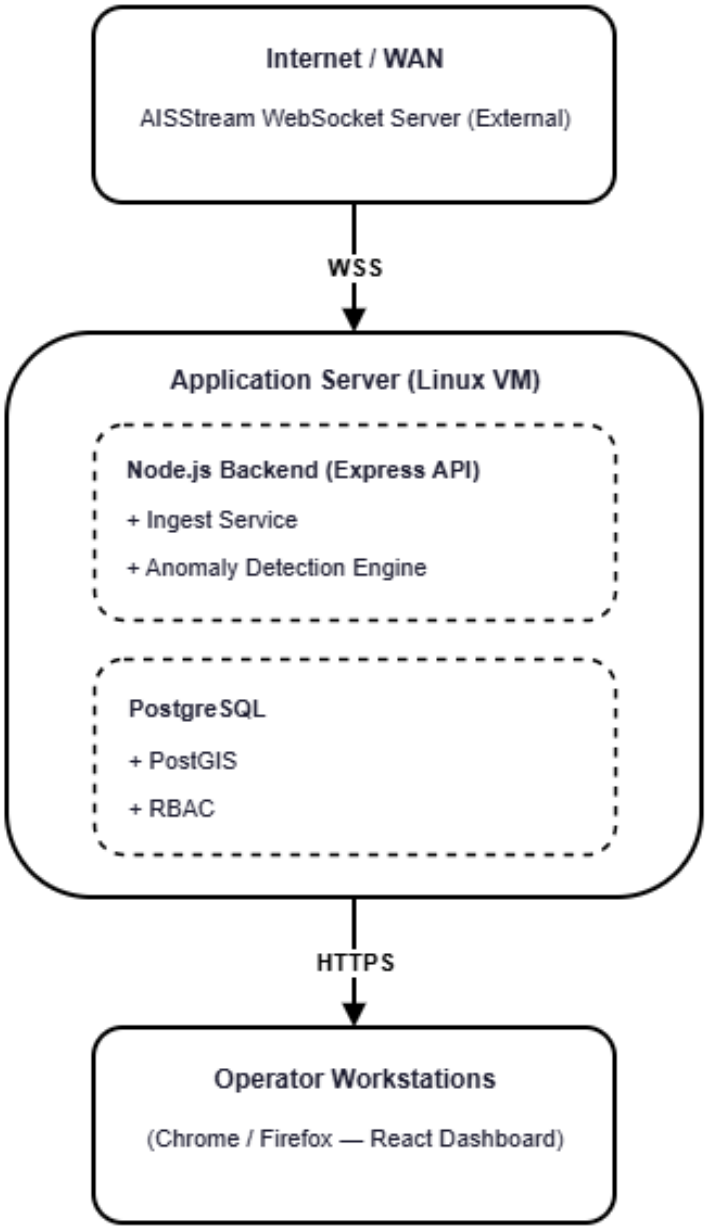


Figure 12: SerenGuard Physical Architecture

Appendix A: Glossary

Term/Acronym	Definition
AIS	Automatic Identification System - a transponder-based system used by vessels to broadcast position, identity, speed, and heading
EO/IR	Electro-Optical/Infrared - sensor technology for visual and thermal imaging of vessels
MMSI	Maritime Mobile Service Identity - a unique 9-digit number identifying a vessel's radio equipment
SAR	Synthetic Aperture Radar - satellite-based radar imaging system
RBAC	Role-Based Access Control - access management based on user roles
Loitering	A vessel remaining in an area for an extended period without an apparent legitimate purpose
Dark Vessel	A vessel that has turned off its AIS transponder to avoid detection
VTS	Vessel Traffic Service - a maritime traffic monitoring service
PostGIS	A spatial database extension for PostgreSQL
JWT	JSON Web Token - a standard for secure authentication tokens
API	Application Programming Interface
REST	Representational State Transfer - an architectural style for APIs
MMDA	Maritime and Marine Domain Awareness

Appendix B: Analysis Models

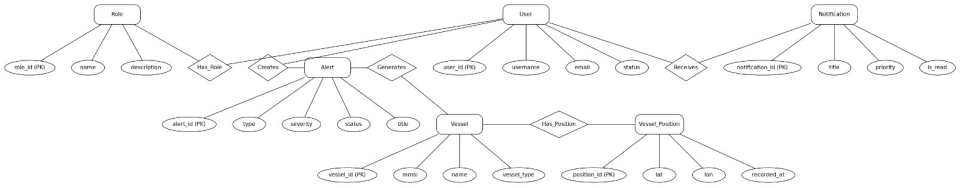


Figure 13: ER Diagram

Appendix C: To Be Determined List

ID	Description	Owner	Target Date
TBD-001	Confirm whether MFA is required	Dr. Chathura Seneviratne	
TBD-002	Confirm minimum 99.9% uptime SLA with client	Dr. Chathura Seneviratne	
TBD-003	Define geofence zones for Sri Lankan waters	Dr. Seneviratne / Team	
TBD-004	Confirm EO/IR sensor hardware specifications for Phase 2	Client	
TBD-005	Finalise Isolation Forest hyperparameters after training		
TBD-006	Define per-vessel-type statistical speed thresholds		
TBD-007	Confirm data retention policy (currently assumed 90 days)	–	